IoT have given way to many security threats, including cyber-attacks. Industrial security systems may include fire alarms, chemical sensors, access control systems, video surveillance units and intrusion detection systems, for a complete solution for protecting workers and their assets.

Biometric access control systems. These are fingerprint access and time attendance control systems mostly found in industries, commercial establishments and offices. These use fingerprints to first register in the database for authentication.

Proximity access control systems. RFID-based proximity access systems are normally used in offices, factories, banks and so on. These are inexpensive, quick and easy to use for door and gate entry systems. Sometimes these are even more effective than video surveillance.

Chemical sensors. An array of chemical sensors is used to detect organic compounds present in gases. Some of these sensors are used in homeland security, analysis, radio frequency detection, sensing toxic industrial materials, bomb detection, toxic vapours and chemical agent simulants.

Magnetic sensors. These sensors are used in many security and military systems. Traditional sensors are complemented by new sensor types such as anisotropic magneto resistor (AMR), giant magneto-resistance (GMR) and giant magneto-impedance (GMI) sensors. These are used for the detection of ferromagnetic and conducting objects, navigation, position tracking and in anti-theft systems.

Fig. 5: Panic alarm system (Credit: www.thehomesecuritysuperstore.com)

Some Leading Manufacturers of Sensors For Security Applications

- Armet Alarm & Electronics
- Bosch Security and Safety Systems
- CP PLUS
- Godrej Security Solutions
- Honeywell
- Infineon Technologies AG
- Kahlenberg Industries
- Knightscope
- National Electronic Security
- Phoenix LiDAR Systems
- The Home Security Superstore
- Tyco International
- Velodyne LiDAR
- YellowScan
- Zicom

Autonomous security robots. Knightscope K5 is a 1.8m (6-feet)-tall autonomous security robot, packed with sensors like lidar array and cameras. It moves about parking lot aisles, hallways/offices, stadiums and shopping malls on the prowl for suspicious activity. It can differentiate between a harmless passerby and potential criminal, and feed all that data to the cloud.

K5 is not meant to replace security guards. It is a fully-autonomous security data machine to fill blind spots. This robot can be deployed where human guards are not safe to patrol, including under dangerous bridges, crime-ridden public parking lots and camps.

Honeywell manufactures physical security solutions for industrial facilities with integrated seamless user control systems.

Kahlenberg Industries is a leading manufacturer of sound signalling products consisting of pneumatic air and electric horns for marine, industrial and mass notification use.

Fig. 6: Knightscope robot (Credit: www.digitaltrends.com)

Cyber security and AI

Smartphones, tablets, personal computers, the Internet and smart TVs are now quite common, especially in big cities. Smart or Internet-connected TVs can recognise your face, listen to your voice and send information to third parties.

A smartphone can track your movements easily through signals. It is possible that retail stores track your every movement, gather data and monitor your behaviour by following Wi-Fi signals from your smartphone. Every time you log on to a computer, there could be a breach of privacy and confidentiality. Chances are that someone could be collecting your sensitive information.

Data brokering is an emerging industry that collects, analyses and sells personal information without the owners' knowledge. Artificial intelligence (AI) packed with sensors and machine learning (ML) could be used in cyber security to minimise data theft and other malicious activities.

To sum up

An electronic sensor system extends its application to various fields such as national defence, home, personal safety, office and industry. As crime rates increase, we feel unsafe unless we have appropriate electronic security systems in place.

Electronic sensors play an important role in security systems. While many of the present sensor technologies have greatly benefited society, these may not be sufficient because of evolving modern lifestyles and rapid technology developments. Some sensing technologies in security applications could be enhanced using robots and AI, especially in cybersecurity and privacy applications. EFY